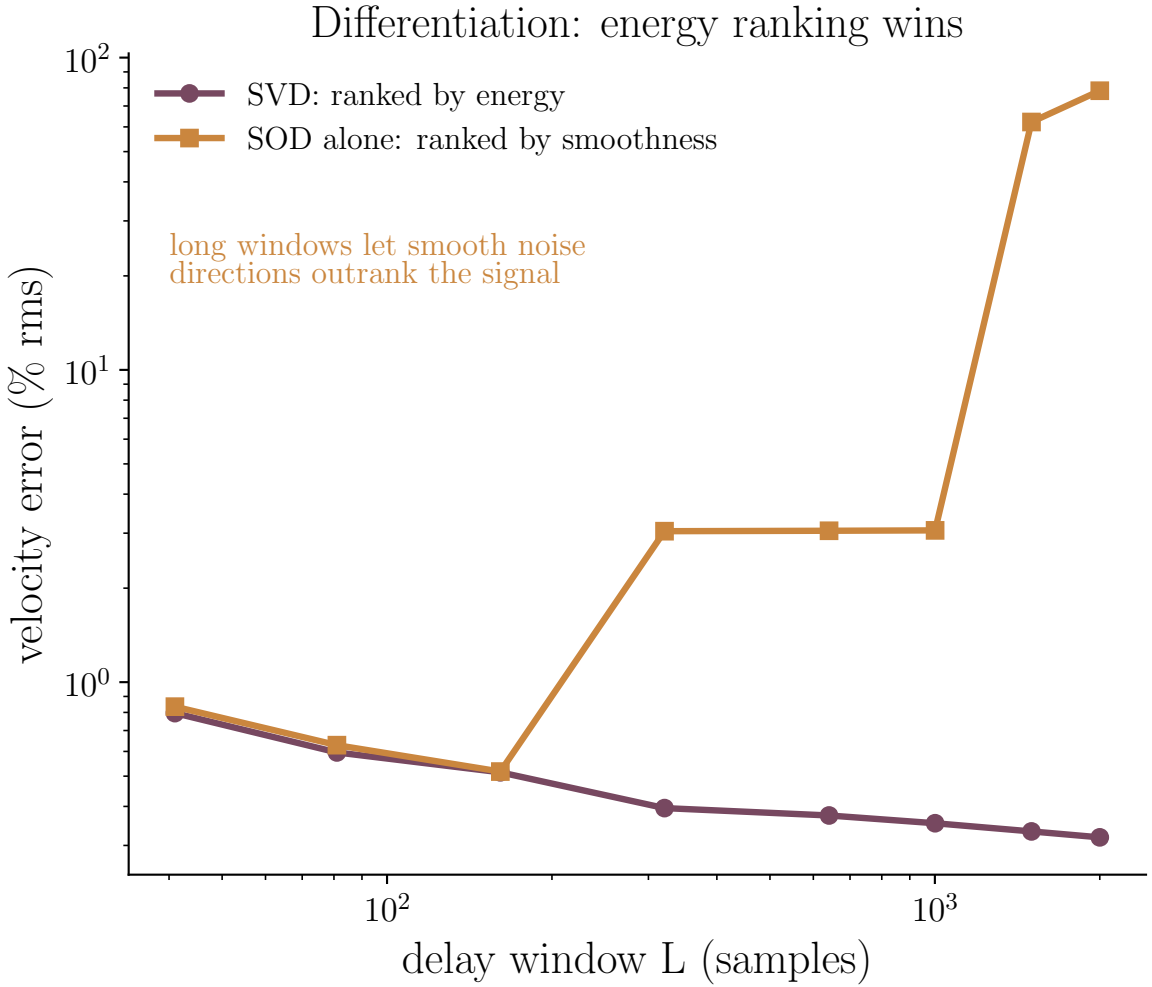
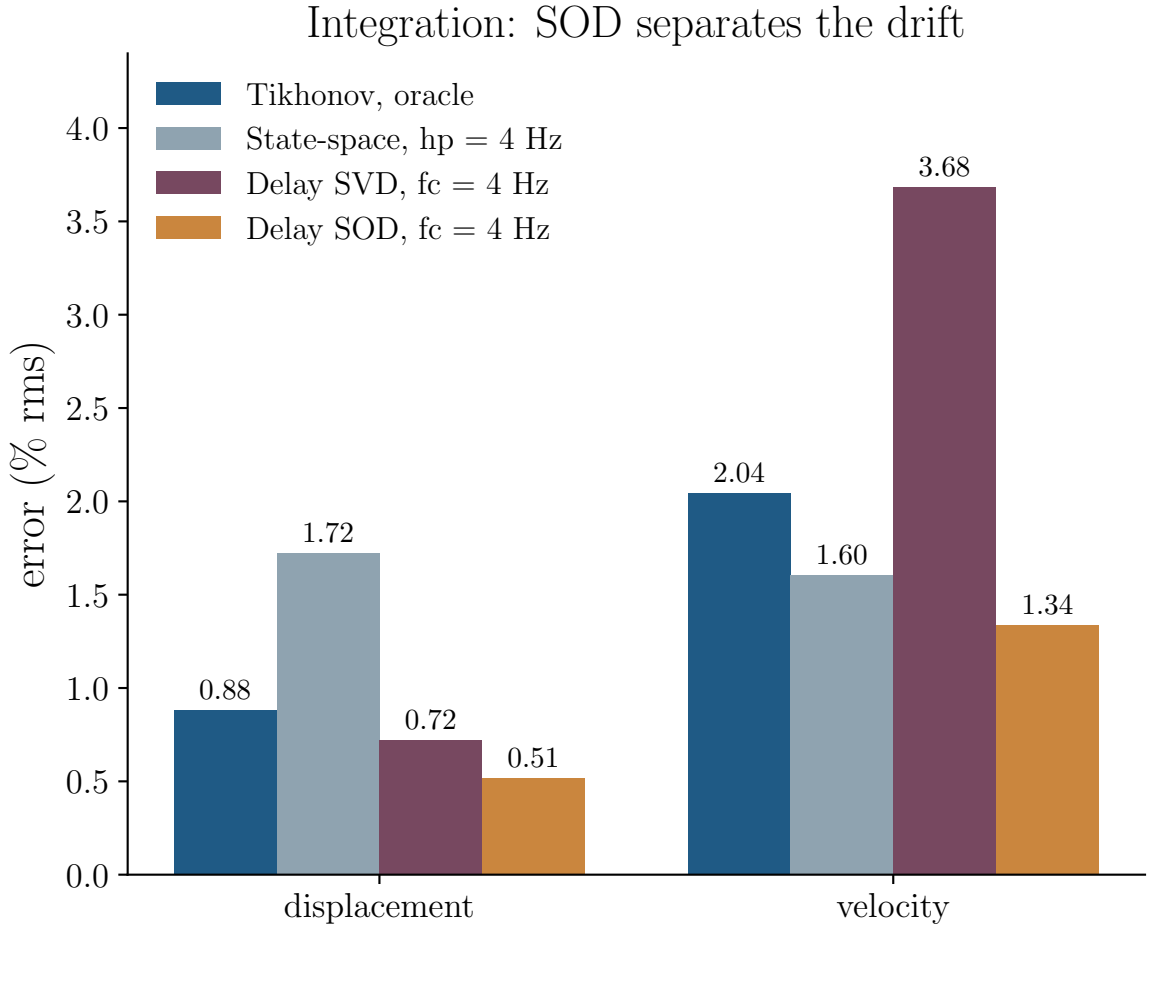

Differentiation: energy ranking wins
SVD: ranked by energy
SOD alone: ranked by smoothness
long windows let smooth noise
directions outrank the signal
velocity error (% rms)
delay window L (samples)


Integration: SOD separates the drift
Tikhonov, oracle
State-space, hp = 4 Hz
Delay SVD, fc = 4 Hz
Delay SOD, fc = 4 Hz
error (% rms)
0.88
1.72
0.72
0.51
2.04
1.60
3.68
1.34
displacement
velocity